JALA Academy
Assignment 3

Loops & Control Flow

Python solution reference • Code examples are organized by the assignment questions.

How to use this document

Save the code examples into a .py file and run with Python 3. Where a program asks for input, enter values when prompted. For file-handling exercises, create the named text file in the same folder first.

1. Print message 10 times

```
for _ in range(10):
    print("Bright IT Career")
```

2. While loop from 1 to 20

```
n = 1
while n <= 20:
    print(n)
    n += 1
```

3. Equal and not equal

```
a = int(input("First number: "))
b = int(input("Second number: "))
print("Equal:", a == b)
print("Not equal:", a != b)
```

4. Odd and even numbers from 1 to 50

```
print("Even numbers:")
for n in range(1, 51):
    if n % 2 == 0:
        print(n, end=" ")
print("\nOdd numbers:")
for n in range(1, 51):
    if n % 2 != 0:
        print(n, end=" ")
print()
```

5. Largest of three

```
a, b, c = 12, 25, 18
print("Largest:", max(a, b, c))
```

6. Even numbers between 10 and 20 using while

```
n = 10
while n <= 20:
    if n % 2 == 0:
        print(n)
    n += 1
```

7. Armstrong number

```
number = int(input("Enter a non-negative integer: "))
digits = str(number)
power = len(digits)
total = sum(int(digit) ** power for digit in digits)
print("Armstrong number" if total == number else "Not an Armstrong number")
```

8. Prime number

```
number = int(input("Enter an integer: "))
```

```
is_prime = number >= 2
for divisor in range(2, int(number ** 0.5) + 1):
    if number % divisor == 0:
        is_prime = False
        break
print("Prime" if is_prime else "Not prime")
```

9. Palindrome number

```
number = input("Enter a number: ")
print("Palindrome" if number == number[::-1] else "Not a palindrome")
```

10. Even or odd

```
number = int(input("Enter a number: "))
print("Even" if number % 2 == 0 else "Odd")
```

11. Gender identification

```
gender = input("Enter M or F: ").strip().upper()
if gender == "M":
    print("Male")
elif gender == "F":
    print("Female")
else:
    print("Invalid Input")
```

12. Multiplication table

```
number = int(input("Enter a number: "))
for i in range(1, 11):
    print(f"{number} x {i} = {number * i}")
```

13. Count digits

```
number = abs(int(input("Enter an integer: ")))
count = 1 if number == 0 else 0
while number > 0:
    count += 1
    number //= 10
print("Digit count:", count)
```